Albanian AI — Project Plan

Oct 11, 2026 · @Giulio

Overview

The goal is to build the first serious open benchmark for AI in Albanian, measure how well today's chatbots and open models handle it, then release a small fine-tuned model that beats comparable open models on practical tasks. It's solo work over roughly 9 months, run alongside the MSc.

The problem: Albanian has around 7 million speakers across Albania, Kosovo, North Macedonia, Montenegro and the diaspora (including Italy, Greece, the UK, Germany, Switzerland and the US). It's a language isolate within Indo-European, with two main dialects (Gheg and Tosk) and rich grammar. Large AI models are trained mostly on English, so Albanian answers are often grammatically off, culturally wrong or mixed with Serbian, Italian or Greek words.

What you deliver:

1. **AlbEval** (working name): an open benchmark of native-written Albanian tasks covering grammar, reading comprehension, culture and facts, public services and law, and translation.
2. **A leaderboard:** results for the main chatbots (ChatGPT, Claude, Gemini) and open models (Llama, Qwen, Gemma, Mistral), published on Hugging Face.
3. **A fine-tuned open model:** a small model (about 3–9B parameters) adapted to Albanian with LoRA, released openly.
4. **A paper:** submitted to a multilingual or low-resource NLP venue.
5. **A demo:** a simple web app showing the model on one useful task, such as explaining official letters or answering questions about Albanian public services.

Why you: few people combine ML training, an AI MSc in progress and fluent Albanian, and you can also check Italian–Albanian translation directly.

Research to do first

The gap is real but not empty: Albanian sits inside a few large multilingual benchmarks and has some small community models, but there's still no broad, native-written Albanian benchmark with a public leaderboard. Spend the first 3–4 weeks confirming that before you build.

What already exists (checked October 2026)

Resource	What it is	Albanian coverage	Why it matters to you
AlbanianLLMSafety	Safety prompt set, May 2026	2,951 prompts, 11 harm categories	Closest prior work; covers safety only and is shared on request. Cite it, don't duplicate it
INCLUDE	Exam questions in 44 languages (ICLR 2025)	576 Albanian questions	Real Albanian exam questions, but few of them. Use as an extra test set
Belebele	Reading comprehension, 122 language variants	900 questions (`als_Latn`)	Translated from English, so not culturally Albanian. Good as a baseline task
FLORES-200	Machine translation test set	Includes `als_Latn`	Standard for measuring translation quality
HPLT 2.0 / FineWeb-2	Large cleaned web text	Albanian web text subsets	Raw text for continued pre-training, if you go that far
Bleta and similar community models	Gemma-based Albanian fine-tunes	No published evaluation	Your benchmark can score these too; possible collaborators

Questions to answer in your literature review

1. Which multilingual benchmarks include Albanian, and were they written natively or translated? Search the awesome-multilingual-llm-benchmarks list, ACL Anthology, arXiv and Hugging Face for "Albanian", "shqip", `sq` and `als_Latn`.
2. What Albanian NLP datasets exist for older tasks (named entities, sentiment, hate speech, part-of-speech tagging)? These can become benchmark tasks.
3. Which Albanian models exist on Hugging Face, and do they report any scores?
4. How do other low-resource benchmarks do it? Read papers on native benchmarks for languages like Basque, Estonian, Greek or Serbian, and copy their method.

5. Which Albanian sources have reusable licences: Matura exam papers, Wikipedia, government sites (such as e-Albania and Kosovo's e-Kosova), public-domain books and news archives?
6. How should you handle Gheg vs Tosk, and Albanian written without ë and ç (very common online)? These are good, distinctive test cases.
7. Who works on Albanian NLP? Check the University of Tirana, the University of Prishtina, and authors of the papers above. A co-author or reviewer from there adds credibility.

Tools for the review: Google Scholar, Semantic Scholar and Connected Papers for finding papers, Zotero for references, and a simple spreadsheet logging each resource (name, link, task, size, licence, native or translated).

Scope and phases

The benchmark is the core product. The model and demo come after it, and can be cut if time runs short without losing the paper.

Phase 1 — Benchmark (AlbEval). Aim for about 2,000–3,000 items across six tasks. Quality matters more than size: every item should be native-written or checked by you.

Task	What it tests	Target items	Where items come from
Grammar and spelling	Cases, verb forms, definite articles, ë/ç restoration	400	Written by you; grammar textbooks; common online errors
Reading comprehension	Understanding real Albanian text	400	Albanian Wikipedia and news, questions written natively
Culture and knowledge	History, geography, literature, traditions of Albania and Kosovo	500	Matura exam papers (check licence); your own questions
Public services and law	Questions about documents, procedures, rights	300	Official government guidance pages
Translation	Albanian to and from English and Italian	400 sentence pairs	FLORES-200 plus your own Italian–Albanian pairs

Task	What it tests	Target items	Where items come from
Open-ended writing	Fluent, natural Albanian answers	200 prompts	Scored by native reviewers and an LLM judge

Phase 2 — Evaluation. Run every task on 3 closed chatbots (via their APIs) and 6–8 open models, including existing Albanian fine-tunes. Report accuracy per task, translation scores (chrF++, COMET), and human ratings on a sample. The analysis is where the paper's findings come from: dialect gaps, missing diacritics, cultural errors, code-switching into other languages.

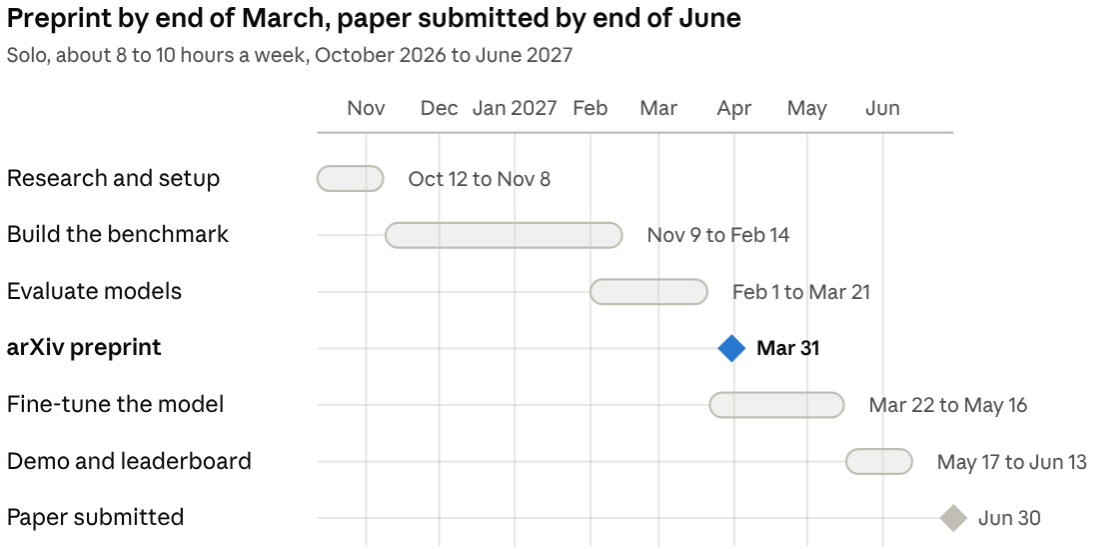
Phase 3 — Fine-tuned model. Take the best small open model from Phase 2 and adapt it with LoRA on Albanian instruction data. Optionally run continued pre-training on Albanian web text first. Never train on the benchmark's test items.

Phase 4 — Demo and release. A Gradio app on Hugging Face Spaces, the dataset with a dataset card, the model with a model card, a public leaderboard, and a GitHub repo.

Out of scope for now: speech, Gheg-only resources, and training a model from scratch. Note them as future work in the paper.

Timeline

The plan takes about 9 months at 8–10 hours a week: a public preprint by the end of March 2027 and the paper submitted by the end of June 2027, before the MSc dissertation takes over the summer.



AlbEval plan · 5 phases, 2 milestones

The benchmark gets the longest stretch because it runs through the Christmas break and January deadlines. If anything slips, shorten fine-tuning rather than move the preprint.

Checkpoints

1. **Early November:** resource spreadsheet done, annotation guidelines written, 60 trial items checked by a reviewer.
2. **Mid-February:** about 2,000–3,000 reviewed items, a hidden test split and a dataset card.
3. **Late March:** all models scored, findings written up, preprint and dataset public.
4. **Mid-May:** a fine-tuned model that beats its base model on AlbEval, released with a model card.
5. **Late June:** demo live, leaderboard public, paper submitted to a workshop or ACL Rolling Review.

Resources

The whole project can run for under about £150 in total; the main cost is your time (about 8–10 hours a week).

Resource	What you need	Free or cheap option	Rough cost
Your computer	16 GB RAM minimum; 32 GB and an NVIDIA GPU (8 GB+ VRAM) helps	Run small quantised models locally with Ollama or LM Studio	£0
GPU for fine-tuning	One GPU with 16–24 GB VRAM for a few hours per run	Kaggle notebooks and Google Colab free tiers; ask about King's HPC access through your department	£0–50
Chatbot API calls	Running ~3,000 items through 3 closed models, plus an LLM judge	Use smaller model tiers for drafts; apply for research credit programmes	Roughly £20–100
Hosting	Dataset, model, leaderboard and demo	Hugging Face (free) and GitHub (free)	£0
Annotation	Writing and checking items	Label Studio or Argilla (both free, self-hosted or on Hugging Face)	£0

Resource	What you need	Free or cheap option	Rough cost
Native reviewers	2–3 people to double-check a sample of items and rate answers	Family, friends, Albanian student societies in London, contacts in Tirana and Prishtina	£0, or small vouchers

People to contact

- **An academic adviser:** your MSc supervisor or personal tutor at King's, or the co-author from your VerifAI paper. A senior name on the paper helps acceptance.
- **Albanian NLP researchers:** authors of the papers in the research table, and staff at the Universities of Tirana and Prishtina.
- **Community model builders:** the people behind Bleta and other Albanian models on Hugging Face, who may share data or help test.
- **Diaspora organisations:** Albanian cultural and student groups in the UK and Italy, for reviewers and early users.

Skills to brush up: evaluation design (how to write fair, unambiguous multiple-choice questions), basic statistics (confidence intervals, inter-annotator agreement with Cohen's kappa), and LoRA fine-tuning.

Tech stack

Everything is Python 3.11+, managed with `uv` (or conda), with code on GitHub. No other language is needed except a little HTML if you build a custom leaderboard page.

Phase	Library or tool	Used for
Data collection	`requests`, `beautifulsoup4`, `trafilatura`	Downloading and cleaning web pages
Data collection	`pdfplumber`, `pymupdf`	Extracting questions from exam PDFs
Data collection	`datasets` (Hugging Face)	Loading INCLUDE, Belebele, FLORES, HPLT; publishing your dataset
Data cleaning	`pandas`, `polars`	Tables of items
Data cleaning	`fasttext` language ID or `GlotLID`	Filtering out non-Albanian text
Data cleaning	`datasketch` (MinHash)	Removing near-duplicate items

Phase	Library or tool	Used for
Annotation	`argilla` or Label Studio	Writing, reviewing and rating items
Annotation	`pydantic`	A strict schema for every benchmark item
Evaluation	`lm-evaluation-harness` (EleutherAI)	Standard runner for open models; add your tasks as YAML configs
Evaluation	`litellm`	One interface to call the OpenAI, Anthropic and Google APIs
Evaluation	`sacrebleu` (chrF++), `unbabel-comet`	Translation scores
Evaluation	`scikit-learn`, `scipy`, `krippendorff`	Accuracy, bootstrap confidence intervals, annotator agreement
Local inference	`ollama`, `vllm`, `llama-cpp-python`	Running open models on your machine or a rented GPU
Fine-tuning	`transformers`, `peft`, `trl`, `bitsandbytes`	LoRA and QLoRA training
Fine-tuning	`unsloth`	Faster, lower-memory fine-tuning on a single GPU
Fine-tuning	`wandb` or `mlflow`	Tracking experiments
Analysis	`matplotlib`, `seaborn`, `plotly`	Paper figures
Demo	`gradio`	Web demo and leaderboard on Hugging Face Spaces
Quality	`pytest`, `ruff`, `pre-commit`	Tests and tidy code
Writing	Overleaf (LaTeX, ACL template), Zotero	The paper and references

Starter install (core libraries only; add the rest per phase):

```
uv venv && source .venv/bin/activate
uv pip install datasets pandas pydantic litellm sacrebleu scikit-learn
transformers peft trl gradio
```

Folder structure

One GitHub repository keeps code, configs and the paper together. Large data and model weights live on Hugging Face, not in Git.

```
albeval/
├── README.md                # what it is, how to run, results table
├── LICENSE                  # code licence (e.g. Apache-2.0)
├── pyproject.toml           # dependencies (uv)
├── .env.example             # API key names only, never real keys
├── data/
│   ├── raw/                 # downloaded sources, untouched (git-ignored)
│   ├── interim/             # cleaned but not yet reviewed
│   ├── benchmark/           # final items, one JSONL file per task
│   │   ├── grammar.jsonl
│   │   ├── reading.jsonl
│   │   ├── culture.jsonl
│   │   ├── public_services.jsonl
│   │   ├── translation.jsonl
│   │   └── writing.jsonl
│   ├── train/               # fine-tuning data, kept apart from the
benchmark
│   └── SOURCES.md           # every source, URL, licence, date collected
├── src/albeval/
│   ├── schema.py            # pydantic model for one benchmark item
│   ├── collect/             # scrapers and PDF extractors
│   ├── clean/               # language ID, dedup, normalisation
│   ├── evaluate/            # runners, metrics, LLM judge
│   └── utils.py
├── tasks/                   # lm-evaluation-harness YAML task configs
├── training/
│   ├── configs/             # LoRA hyperparameters per run
│   └── train_lora.py
├── notebooks/               # exploration only, never the source of truth
├── results/
│   ├── raw_outputs/         # every model answer, saved for re-scoring
│   └── scores.csv
├── app/                     # Gradio demo and leaderboard
│   └── app.py
├── paper/                   # LaTeX source and figures
├── docs/
│   ├── annotation_guidelines.md
│   └── dataset_card.md
└── tests/
```

One benchmark item (one line in a JSONL file):

```
{"id": "culture-0001", "task": "culture", "question": "...", "choices":
["...", "...", "...", "..."], "answer": 2, "dialect": "standard", "source":
"original", "author": "GD", "reviewed_by": ["R1"], "licence": "CC-BY-4.0"}
```

Save every raw model output. Then you can re-score with a new metric without paying for the API calls again.

Publishing, career and money

Publish the benchmark first and fast: an arXiv preprint plus the Hugging Face dataset makes you the reference point for Albanian AI evaluation before anyone else claims it.

Where to publish (check each venue's current deadlines; most ACL venues go through ACL Rolling Review, which runs roughly every two months):

- **Workshops (realistic first target):** MRL (Multilingual Representation Learning), LoResLM, LoResMT, VarDial (dialects, good for Gheg vs Tosk), SIGTYP.
- **Main venues (stretch):** ACL, EMNLP or NAACL main or Findings, LREC (biennial), or the NeurIPS Datasets and Benchmarks track.
- **Preprint:** arXiv, the moment the first version of the benchmark and results are ready.

Career value

- A first-author paper plus a cited dataset is strong for PhD, Cambridge MPhil or Oxford applications, and for AI lab roles in multilingual work, evaluation or safety.
- A public leaderboard gets you noticed by labs, which routinely look for evaluation data in under-served languages.
- It gives you a talk topic for meetups, King's AI societies and Albanian diaspora events, and strong content for your personal website.

Money routes (indirect at first; don't expect income before the paper is out)

1. **Evaluation consulting:** companies launching AI products in the Balkans need to know how well they work in Albanian.
2. **Data and annotation contracts:** AI labs and data vendors pay for high-quality native data in low-resource languages.
3. **Localisation and translation services:** your translation background plus the model makes an Albanian–Italian–English AI-assisted service.
4. **Public sector and NGOs:** digital government projects in Albania and Kosovo, EU-funded programmes and diaspora organisations.
5. **Grants:** research and open-source grants for low-resource languages, and the King's Entrepreneurship Institute if you turn the model into a product.

6. **A product later:** an Albanian assistant for one job (official letters, schoolwork help, business writing) with a free tier and paid plan.

Risks, ethics and licensing

The biggest risks are scope creep and leaked test data; both are solved by finishing a small, clean benchmark before touching the model.

Risk	What could happen	How to prevent it
Someone publishes first	Another Albanian benchmark appears mid-project	Post a preprint early; focus on native-written and culture-specific items, which are hard to copy
Copyright	Exam papers or news text can't be redistributed	Log every source's licence in `SOURCES.md`; write questions in your own words; ask the publisher when unsure
Data contamination	Test items end up in a model's training data, making scores meaningless	Keep a hidden test split; don't post full test items in plain text on public pages; add a canary string
One-person bias	Items reflect only your dialect, region or views	Use 2–3 reviewers from Albania and Kosovo; report agreement scores
Sensitive topics	Questions on history, borders or religion read as political	Cover facts, not opinions; have a reviewer from each country check them
Personal data	Scraped text contains names or contact details	Remove personal data during cleaning; don't use social media posts about private people
Model and API terms	Some providers restrict using their outputs to train other models	Read the terms before using any chatbot output as training data
Time with the MSc	Coursework and the dissertation squeeze the project	Fixed weekly hours; cut Phase 3 or 4 before cutting benchmark quality

Licences to choose: CC-BY-4.0 for the benchmark (or CC-BY-SA if a source requires it), Apache-2.0 for the code, and the base model's own licence for the fine-tuned model.

Ethics approval: if reviewers are paid or you collect their personal details, check whether King's requires ethics review. Ask your supervisor early, as it can take weeks.

Next steps this week

- [ ] Create the GitHub repo `albeval` with the folder structure above, and a Hugging Face account or organisation
- [ ] Download INCLUDE (Albanian), Belebele (`als_Latn`) and FLORES-200 Albanian, and look through 50 items of each
- [ ] Run 20 Albanian questions through ChatGPT, Claude, Gemini and one local model via Ollama; note the error types you see
- [ ] Read the AlbanianLLMSafety paper and two other low-resource benchmark papers; start the resource spreadsheet
- [ ] Write 30 grammar questions and 30 culture questions yourself to test the item format
- [ ] Draft one page of annotation guidelines (what a good item looks like, how to mark the answer, dialect labels)
- [ ] Email your supervisor or tutor about the project, a possible academic adviser, and ethics requirements
- [ ] Ask 2–3 Albanian speakers (one from Kosovo) whether they'd review items

Sources

- AlbanianLLMSafety: A Safety Evaluation Dataset for Large Language Models in Albanian (arXiv, May 2026)
- INCLUDE base 44 dataset (Hugging Face)
- Belebele dataset (Hugging Face)
- Bleta-sq-2B model page
- HPLT 2.0 cleaned corpus (Hugging Face)
- awesome-multilingual-llm-benchmarks (GitHub)